Kyrylo Kolesnichenko

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Education

Vilnius University, Institute of Social Sciences and Applied Informatics — Kaunas, Lithuania
B.Sc. Information Systems and Cyber Security, GPA 9.5/10 — *Sep. 2023 – Jan. 2027 (expected)*

Fully funded exchanges: Kent State University, USA (ISEP, Spring 2026, GPA 3.8/4.0); University of Vermont, USA (ISEP, Fall 2025, GPA 3.8/4.0); Tampere University, Finland (Erasmus+, Spring 2025, GPA 4.67/5)

Relevant coursework: Advanced Mathematics, Data Mining, Statistical Modelling, Machine Learning, Algorithms

Experience

Undergraduate Robotics Researcher — Mar. 2026 – Present (part-time, remote since Nov. 2026)
Advanced Telerobotics Research Lab, Kent State University, Advisor: Dr. Jong-Hoon Kim — *Kent, OH, USA*

- Lead K1/G1 humanoid RL and motion tracking from simulation to hardware (SONIC, BeyondMimic, Kimodo); **first author and co-author of robotics research papers**.
- Open-sourced a **video-to-humanoid pipeline** combining monocular motion recovery (GVHMR), GMR retargeting, reinforcement learning in Isaac Lab/BeyondMimic, and MuJoCo validation. Tested on the real K1 in consecutive hardware runs without falls, producing crossover steps narrower than the stock controller's gait.
- Co-authored a controlled Pepper LLM/HRI study of motivational interviewing and empathy (**IHCI 2026** [3]).
- Led a university-wide robot-learning workshop (**13+ participants**) and multiple ATR Lab workshops. Featured in a **Vilnius University article**; designed robot footwear, won a robot fashion hackathon award and represented the lab at Kent State Preview Day (**Fox 8 coverage**).

Software Engineering Intern — Mar. 2025 – May 2025
SoftServe — *Remote*

- Developed user authentication and recommendations end to end with React, Node.js and PostgreSQL.
- Designed relational schemas and REST APIs; collaborated in an agile team using Git, code review and testing.

Publications

[1] **K. Kolesnichenko**, I. S. Cardenas, J.-H. Kim. "Learning Expressive Humanoid Locomotion from Monocular Runway Videos for Robot Fashion Shows." *IEEE/RSJ IROS 2026*, Late Breaking Results (LBR). **Presented.** [arXiv] [code] [video]

[2] **K. Kolesnichenko**, M. Ambraziūnas. "Can Structured Prompting Overcome Knowledge Bias in Vision-Language Models?" *Proc. 31st Int. Conf. on Information Society and University Studies (IVUS 2026)*, Kaunas, CEUR Workshop Proceedings. **Published.** [paper] [code]

[3] R. Ashraf, Y. Liu, **K. Kolesnichenko**, S. Ganji, J.-H. Kim, J. Y. Lee. "Can Motivational Interviewing Guided Dialogue Improve Empathy in Human–Robot Interaction? A Controlled Study with Baseline Comparison." *Int. Conf. on Intelligent Human–Computer Interaction (IHCI) 2026*. **Accepted.**

[4] One paper. *IEEE ICRA 2027*. **Under review.**

[5] **K. Kolesnichenko**, I. S. Cardenas, J.-H. Kim. "SONIC-K1: Porting a Universal Humanoid Motion Tracker to a Small Humanoid on a Single GPU." *arXiv:2610.39327*, 2026. **Preprint.** [arXiv] [code] [weights]

Selected Projects

SONIC-K1: Motion Tracker for the Booster K1 | *NVIDIA SONIC, Isaac Lab, H100, MuJoCo, ONNX* [code] [weights] [page] — Oct. 2026

- First open-weights K1 tracker [5]: SONIC/G1 port, trained from scratch on ∼10,000 retargeted mocap clips in ∼100 GPU-hours on one shared H100 (**under 1%** of SONIC compute). Unseen success (MuJoCo, SONIC rule): **97.4%** of 1,985 held-out clips; **93.6%** external PHUMA; 99.0% 100STYLE. **95.9%** under combined simulated hardware constraints; GPU-free laptop ONNX inference.

LLM-Based Technical Assistant for Robotics Software Development | *Bachelor's thesis* — In progress

- Developing a locally hosted **LLM-based assistant** that combines robotics knowledge retrieval with context-aware generation to support implementation and troubleshooting. Planned evaluation targets technical accuracy and source traceability.

Stylized and Teleoperable Locomotion for the Booster K1 | *AMP, mjlab, PPO, MuJoCo, CAD* [code] — 2026 – Present

- Trained and deployed **multiple locomotion policies**, including stylized catwalks, VR teleop and footwear balance, with SONIC fine-tuning. Simulated falls: **under 2.5%** in teleop and **3.9%** in heels; designed and tested shoes, including heels and clogs.

Technical Skills

Robotics / RL: Isaac Lab/Sim, MuJoCo, mjlab, ROS 2, Nav2, PPO/RSL-RL, AMP, BeyondMimic, SONIC, URDF/USD/MJCF
ML / LLMs: PyTorch, TensorFlow, Hugging Face, vLLM, ONNX, OpenCV, scikit-learn, RAG, FAISS, tool calling, structured outputs
Programming: Python, C++, SQL, JavaScript/TypeScript, Bash, R, LaTeX
Systems / Web: Linux, Git, Docker, CUDA, H100, FastAPI, PostgreSQL, React, Node.js, CAD, 3D printing

Honors

1st place, Best Humanoid Wearable/Accessory ($1,000 team award), Humanoid Robot Fashion Challenge, Kent State Design Innovation (Oct. 2026)
Dean's List, Kent State (Spring 2026) · **Merit Scholarship** ×5, Vilnius University · **Exchange grants:** ISEP ×2, Erasmus+